

Mradul Singh

AI / ML Engineer · LLM Agents, RAG & Distributed Systems

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Summary

- Backend engineer with **~3 years** building **distributed, event-driven systems** at scale — CQRS, Kafka, Temporal, gRPC, and multi-tenant authorization.
- Core contributor to a **Java 21 modular monolith** and sole author of **two greenfield services**; primary owner of the org's **gRPC/Protobuf contracts** across 5+ services.
- Deep consensus-adjacent experience: transactional outbox for exactly-once publishing, replay-safe Temporal workflows, and a Zanzibar-style **OpenFGA** authorization model.
- Extended backend foundations into production **AI/agent systems** — LLM agents, RAG pipelines, and eval harnesses — all running on the same distributed infrastructure.

Experience

- **Software Engineer, Accolade** (Jan'25- Present)
 - Core contributor to a **Java 21 / Spring Boot modular monolith** (reads via API, writes via events): shipped end-to-end **CQRS** domains — property inspections (gRPC lifecycle, immutable versioned templates, Temporal-driven expiration & report generation), work orders (upsert + lifecycle + review-strategy approval), and technician timesheets.
 - Implemented a **Kafka transactional outbox** to eliminate the dual-write problem — domain events written in the same DB transaction, then published to Kafka by **Temporal** polling workflows; later shipped a continuously-running publisher workflow (V2).
 - Built **multi-tenant, replay-safe Temporal context propagation** — a `ContextPropagator` carrying org/user/property IDs deterministically across workflows and activities, with worker-thread cleanup to prevent context leakage and memory leaks.
 - **Primary owner** of the org's contract-first **gRPC/Protobuf** API layer (Buf) shared across 5+ services (500+ commits) — schema versioning, deprecation discipline, and protovalidate rules.
 - Designed and own a Zanzibar-style **ReBAC OpenFGA** authorization model plus its Java client — `listAuthorizedResources`, a global `SecureAspect` authorization toggle, relation-collection service, and authz metrics (5 modules, 11 resource types).
 - Built event-driven **Kafka** domain-event pipelines (lead / work-order / technician events, key-serialization strategies) and a **pgvector**-backed knowledge-base module extracted into standalone Maven modules behind a clean API.
 - Built **two greenfield services solo** — **Helios** (rack-aware Kafka→S3 archival: single-consumer, commit-only-on-flush offsets, per-record DLQ, Snappy protobuf, Temporal-scheduled) and **Optima** (a Timefold work-order-routing solver, 11 constraints, S3/SQS-decoupled with a Valkey driving-time cache).
 - **AI edge**: built an in-product Claude agent (Athena) on Anthropic's Claude Managed Agent API, a **LangGraph**/Bedrock agent backend (iris), and a **Temporal + NVIDIA NeMo** agent-evaluation harness (hera) — all event-driven, gRPC-connected, and OpenTelemetry-traced.
 - Owned supporting platform: AWS infrastructure-as-code (ECS/Fargate, IAM, S3→SQS, Lambda via **Terraform/CDK**) and Databricks (Lakebase, DLT, Unity Catalog); contributed **2,500+ commits / 450+ merged PRs across 15 repositories** in ~16 months.
- **Software Developer, PeakFlo** (Jan'24- Jan'25)
 - Built and optimized core **payment** processing on a TypeScript platform (Firebase, GCP), cutting internal payment-API latency around **50 ms** and revamping a business-critical payment-gateway integration.
 - Led development of a new **travel-&-expense** module — data model, APIs, and user-facing flows — expanding the platform into a new product segment.
 - Integrated **accounting/ERP** systems (QuickBooks/Xero-style sync) to auto-post expenses and payments and cut manual bookkeeping.
 - Built configurable **approval workflows** for expense and payment authorization (multi-level approvers, spend limits, policy checks).
- **Backend Engineer, Ostello AI** (Sep'23- Jan'24)
 - Scaled backend services from **1K to 100K users** by fronting them with an **nginx** reverse proxy using IP-hash load balancing (Node.js/NestJS, GraphQL, PostgreSQL).
 - Cut database connection setup/teardown from **10 ms to 3 ms** by introducing **connection pooling** (pg-pool), improving backend throughput under load.
 - Built a secure authentication system with **multi-factor authentication (MFA)** and email verification.
 - Designed RESTful & **GraphQL** APIs and a MongoDB-backed **event-logging pipeline** (~10K events/hour) for request tracing and analytics.

Key Projects

- **agentctl** — Durable, observable runtime for LLM agents *Python, Postgres, S3, OTel*
 - Crash-resumable execution via a **write-ahead log** + deterministic state machine — interrupted runs resume from the last committed step with **exactly-once, idempotency-keyed tool side-effects**.
 - Durable **human-in-the-loop**: suspend an agent run for hours awaiting approval, then resume it safely without losing context.
 - **OpenTelemetry**-native tracing + **per-tenant token-cost attribution** + **CI eval-gates** that fail the build on quality regressions.
 - Provider-agnostic adapter (Claude / Bedrock / OpenAI) wrapping any agent loop; fault-injection harness that kills the process mid-tool-call.
- **fga-gateway** — ReBAC auth proxy for LLM agents & MCP *Go, OpenFGA, MCP*
 - Enforces per-object, **relationship-based (Zanzibar/OpenFGA)** permissions on every agent tool call and MCP request. An agent can access only what the end user is authorized to.
 - **Policy-as-code** DSL mapping tool-call intents to authorization checks, with safe delegation and a full allow/deny **audit trail**.
 - Deployed as a **sidecar proxy** with sub-millisecond check caching and **fail-closed** defaults; treats retrieved content as untrusted so prompt injection can't bypass authz.
 - **Threat-model design doc** on the agent-authorization boundary.
- **logcask** — Raft-replicated distributed commit log *Java, Raft*
 - Implemented an append-only **segmented-log storage engine** — memory-mapped segments, batched **fsync**, a sparse offset index, and background compaction — exposing an ordered, durable record stream.
 - Wrote a **Raft** consensus layer from scratch: leader election, log replication, snapshotting/log compaction, and dynamic membership changes for strongly-consistent replication across a cluster.
 - Proved correctness with a **Jepsen-style linearizability harness** that injects network partitions and kills the leader mid-write, asserting zero lost or duplicated records across failovers.
 - Built a producer/consumer API with **consumer groups**, offset commits, and **exactly-once** delivery via idempotent producers; added backpressure and configurable durability (acks).
 - Instrumented with **OpenTelemetry** and load-tested sustained throughput/latency.
 - Wrote a **design doc** on the consistency model, replication protocol, and failure modes.

Technical Skills

- **General:** Distributed Systems, Event-Driven Architecture, Microservices, CQRS, System Design, Agentic AI Infrastructure, RAG Systems, Multi-tenancy, ReBAC Authorization, Optimization Systems, Scalable Databases
- **AI / LLM:** Anthropic Claude (Agent SDK, Claude Managed Agent), AWS Bedrock (Converse, Nova, Guardrails), LangGraph, LangChain, LangSmith, RAG & Hybrid Retrieval, LLM-as-judge Evals, Agent Orchestration, MCP, pgvector, NVIDIA NeMo, Prompt & Context Engineering
- **Programming Languages:** Python, Java 21, TypeScript, JavaScript, SQL, Protobuf, C, C++, HTML/CSS
- **Backend & Infrastructure:** Spring Boot, Spring AI, FastAPI, Node.js, Express.js, NestJS, gRPC/Protobuf (Buf), REST & GraphQL API Design, Apache Kafka, Temporal, Timefold Optimization, Docker, Kubernetes, Nginx, Terraform/CDK, CI/CD (GitHub Actions), Reverse Proxying
- **Frontend & Tools:** React.js, Redux Toolkit, TanStack Query, Tailwind CSS, Vite, Figma, Git, GitHub
- **Databases, Cloud & Observability:** PostgreSQL (+ pgvector; schema optimization & tuning), MongoDB, MySQL, Redis/Valkey, Databricks (Lakebase, DLT, Unity Catalog), AWS (ECS/Fargate, Lambda, S3, SQS, Elastic Beanstalk), GCP, Firebase, OpenFGA/Auth0 FGA, OpenTelemetry, Datadog, Grafana/Loki

Education

- B.Tech + M.Tech (Dual Degree) - Indian Institute of Technology (BHU), Varanasi 2021 – 2026

Positions of Responsibility

- **Tech Head** — Technex & Spardha, IIT (BHU) Varanasi 2023 – 2024
 - led technology for the institute's flagship technical and sports festivals.
- **Core Member** — Club of Programmers (COPS), IIT (BHU)

Achievements

- **Competitive Programming:** CodeChef 4-star (max 1897) · Codeforces Pupil (max 1510) · rank 514/30K, Codeforces Round 860 · rank 44/20K, CodeChef Starters 80.
- **Scholastic:** Top 5%ile JEE Main · AIR 2K, AKTU (of 0.12M) · KVPY & PRMO qualifier · Team Rank 3, Flipkart GRiD 4.0 (100+ teams).